Science Academies' Summer Research Fellowship Programme

4-Week Report

Particle Physics: Simulations and Machine learning

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As experimental setups become more complex, simulations and machine learning provide essential tools to understand interactions of particles and data analysis. This report outlines the progress made in the first four weeks of the summer project. It highlights the importance of simulations and data-driven models in studying high-energy physics.

One of the most extensively used simulation platforms in high-energy physics is GEANT4 [1–3]. It is an object-oriented software toolkit used to simulate the passage of particles in matter. They are essential in detector design, optimization, and performance evaluation. It allows the simulation of realistic experimental conditions like particle trajectories, energy deposition and detector responses with high accuracy. In addition, GEANT4 supports a wide range of physical processes, including electromagnetic, hadronic, and optical interactions, making it highly versatile for various experimental needs. These generate synthetic datasets and evaluate reconstruction techniques without performing costly and time-consuming physical experiments.

In these four weeks, I have worked on the construction of detector geometries in GEANT4. A simulation on Rutherford scattering was performed in which a beam of 20,000 alpha particles of energy 5.5 MeV were incident on a gold foil of thickness 2 μm (Fig. 1). It is observed that only 3 alpha particles were backscattered. This shows the rare nature of large-angle scattering in Rutherford's experiment. It provided insights into the interaction of particles with matter.

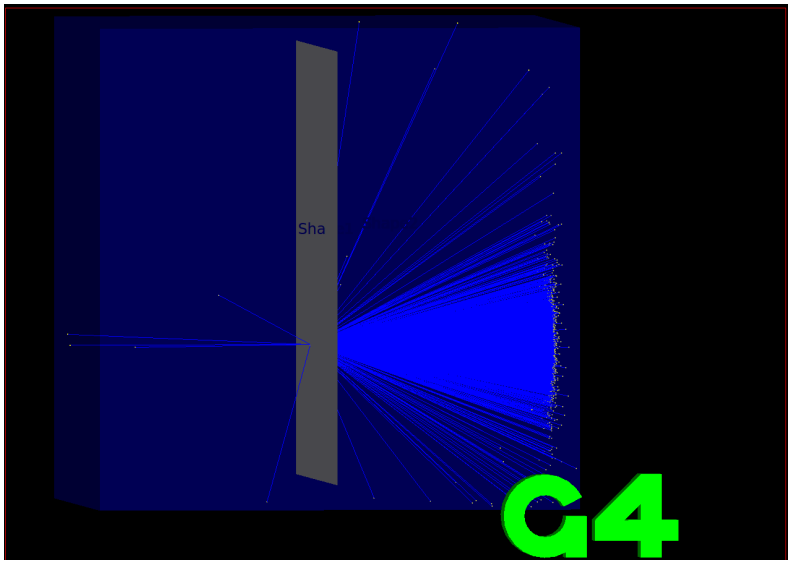


Figure 1: Simulation of alpha particle scattering by Gold foil in GEANT4.

Tracker-type detectors [4] are used to determine the trajectories of charged particles. Technologies like drift tubes, spark chambers, scintillating strips, resistive plate chambers (RPCs) [5], etc., are used to provide good position resolution. For weakly interacting particles like neutrinos, precise reconstruction is required for optimal data interpretation.

Machine Learning [6, 7] is a powerful tool for enhancing data analysis in particle physics. It has various applications in noise suppression, event classification and parameter reconstruction. Data-driven models can supplement traditional analysis methods due to the large amount of data generated by modern detectors. The upcoming phase of the project will focus on learning various ML models. We will train and evaluate the performance of several models using various performance metrics.

References

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